Chapter 1
Data File and Configuring SQL Server

Chapter Outline

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- Files and Filegroups
- Transaction Logs
- Access State
- Configuring SQL Server
- SQL Server Instances

Introduction

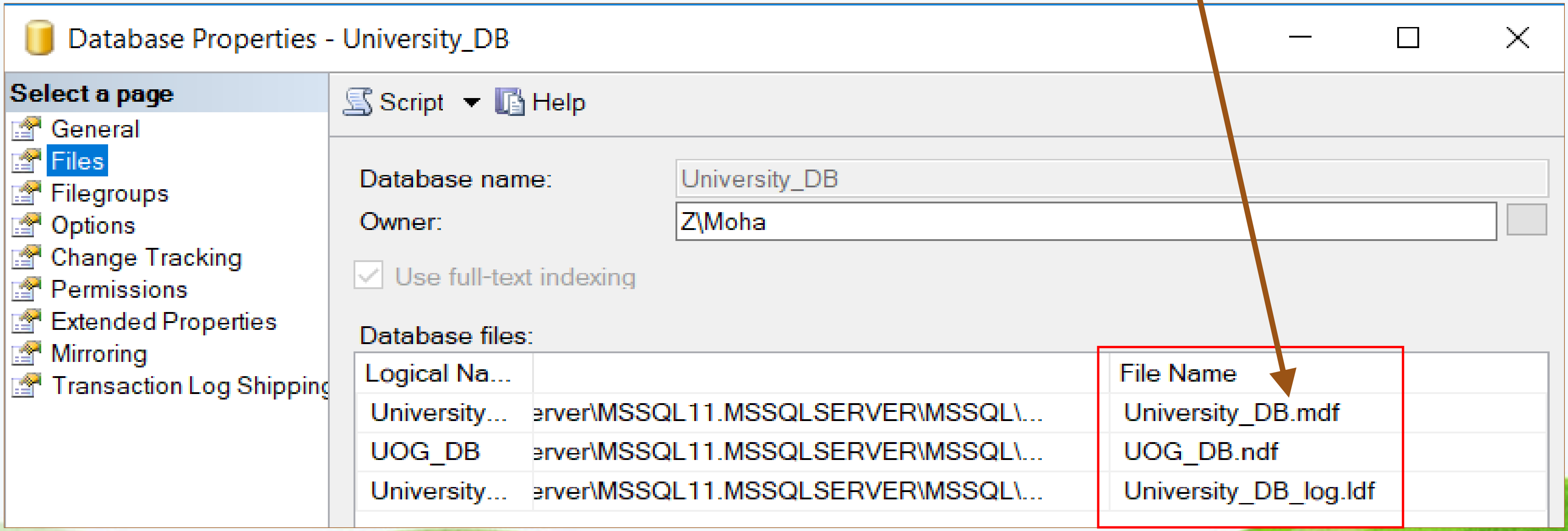
- Data within a database is stored on disk in one or more data files.
- SQL Server databases have two basic types of files:
  - Rows data (primary) files
  - Transaction log
- **Data files** are responsible for the long-term storage of all the database objects.
- Data files can be grouped together in **filegroups** to link physical storage into a database (for allocation) and administration purposes.

Files and Filegroups

- Filegroups are a logical structure, defined within a database, that map a database to the data files on disk.
- Filegroups can contain more than one data file.
- When the database is created, the primary filegroup is marked as the default filegroup.
- Following the initial creation of the database, you can add filegroups as needed to separate the storage of objects within the database.

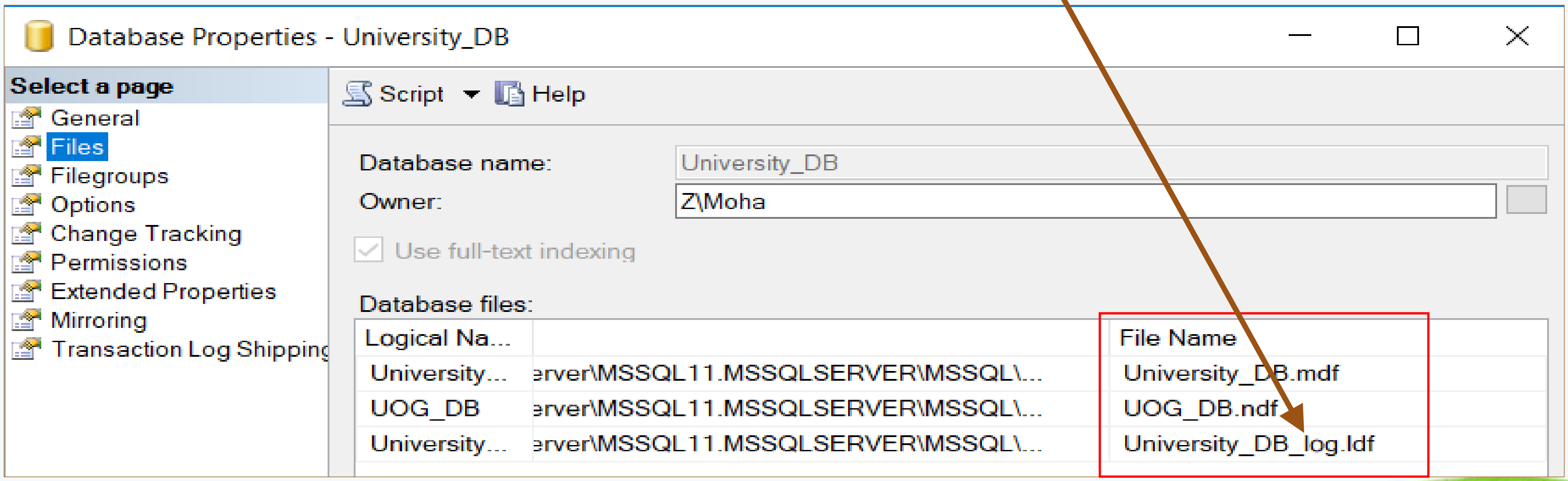
Primary Data File and Filegroup

- A file with **.mdf** extension is the first (**primary**) data file that is created within a database, and is associated with the primary filegroup named PRIMARY which contains all the system objects necessary to a database.



Transaction Log File

- Transaction log files (**.ldf** extension) are responsible for storing all the transactions and changes that are executed against a database.
- Information stored in **log files** is required to recover all transactions in the database.



Access State

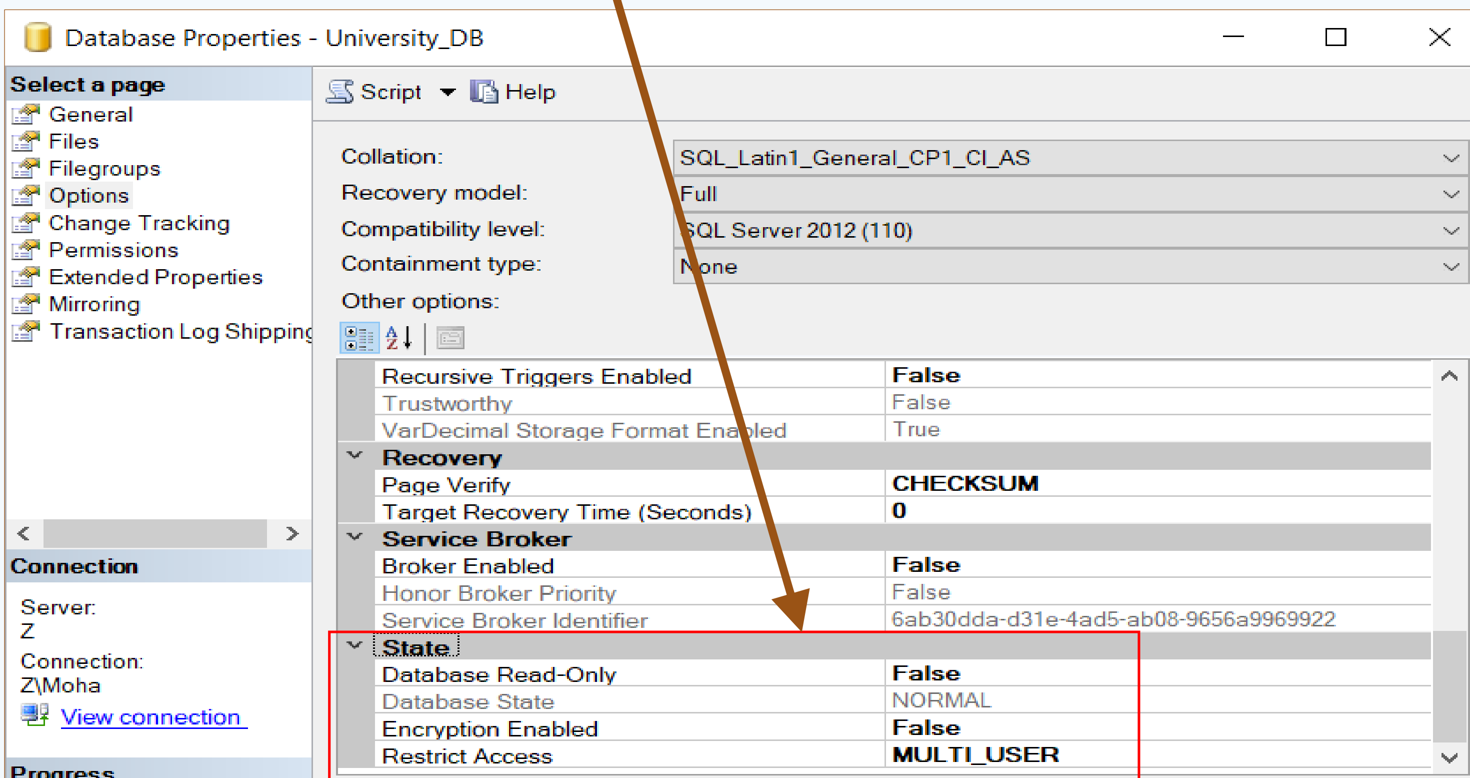
- Access to a database can be controlled through several options.
- The state of a database can be explicitly set to:
  - **ONLINE:** allows you to perform all operations.
  - **OFFLINE:** the database is inaccessible.
  - **EMERGENCY:** only SELECT command is allowed to be executed.
- Control the ability to modify data by setting the database to either **READ_ONLY** or **READ_WRITE**.

Access State (Cont'd…)

- User access to a database can be controlled through the following options.
  - **SINGLE_USER mode:** only a single user session is allowed to access the database.
    - ✓ It is used for maintenance actions.
  - **RESTRICTED_USER mode**: allows only members of the **db_owner**, **dbcreator**, or **sysadmin** roles to use the database.
  - **MULTI_USER mode:** allows multiple users to access the database at once (multiple sessions are allowed).

Access State (Cont'd…)

- The normal operational mode for most production databases is **ONLINE**, **READ_WRITE**, and **MULTI_USER**.



Configuring SQL Server

- After installing MS SQL Server, you can configure various services of the database within the core database engine using a configuration tool known as **SQL Server Configuration Manager**.
- This tool is responsible for:
  - Starting, stopping, restarting a service.
  - Managing the start-up mode (Automatic, Disabled, Manual) of a service.
  - Configuring SQL Server network protocol (TCP/IP).
  - etc…

SQL Server Instances

- SQL Server instances define the container for all operations you perform within SQL Server.
- Each instance contains its own set of databases, security credentials, configuration settings, and other SQL Server objects.
- You can install one instance as the default instance along with multiple additional named instances.

SQL Server Instances (Cont'd…)

- When you connect to a default instance of SQL Server, you use the name of the machine.
- When connecting to a named instance, you use the combination of the machine name and instance name as:
  - machineName\SQLServerInstanceName